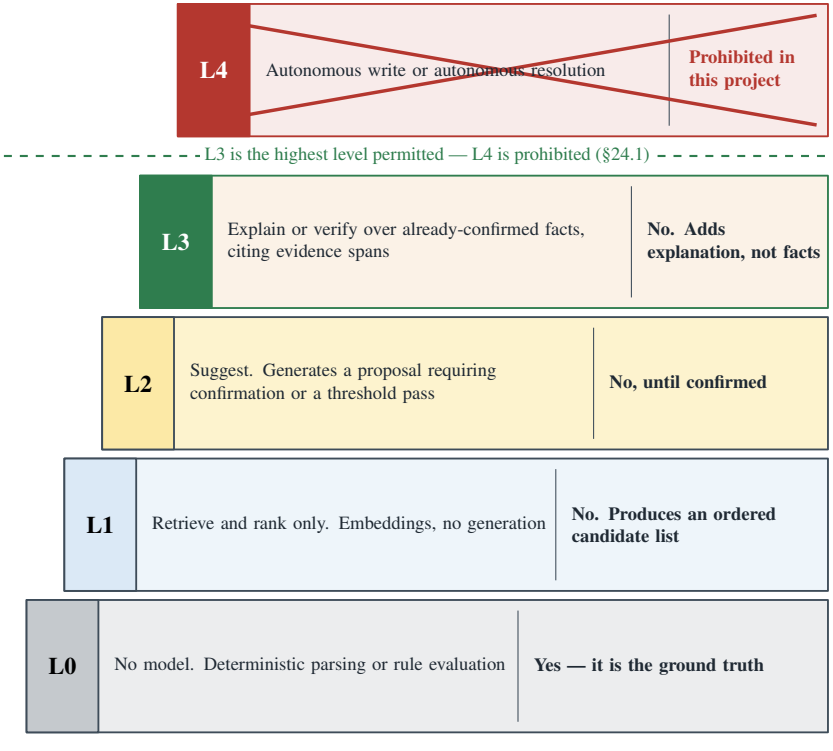
Invariant (§24): no LLM output writes to the graph of truth; no LLM output fires a policy on its own.

Autonomy ladder (§24.1) — autonomy vs. system state



Usage register (§24.2) — every stage has a deterministic fallback

stage	what the LLM may do	rung	deterministic fallback
Ingestion	Extract symbols, signatures, docstrings, comments, imports; tree-sitter parsing	**L0**	None needed — deterministic
Candidate generation	Propose requirement-to-code candidates; hybrid BM25 and code embeddings, fused by reciprocal rank	**L1**	Lexical retrieval alone
Adjudication	Decide whether a candidate is a real trace link; local instruct model over a context packet, constrained JSON verdict with cited spans	**L2**	Similarity threshold only, reported as a baseline
Drift review	Does this anchor still hold after this diff? The model reads the diff hunk and the anchored symbol	**L2**	Symbol-name and content-hash re-anchoring only
Conflict explanation	Render why a policy fired; template filled from the evidence path, the model may rephrase, never add facts	**L3**	Raw template text
Adaptation	LoRA fine-tune for verdict format and domain vocabulary; committed work, not a contingency	**L2**	Base model with few-shot prompting
Verdicting	Structured criterion-level verdict with citations over one immutable packet; deterministic verdict gate before any write (§11.8)	**L2 + gate**	Deterministic Rules 1–3 plus unresolved-evidence reporting; verification marked unavailable (§14)

Chain summarization — removed from committed scope (§24.2) · the model is bounded, not central: one structured verdict with citations over one immutable packet; it does not choose evidence, create edges or fire alerts (§11.2).

Degraded mode (§14, §24.2): with the model server switched off the deterministic paths survive — tree-sitter ingestion (L0), lexical retrieval alone (L1), symbol-name/content-hash re-anchoring, raw template text and deterministic Rules 1–3. L2/L3 inference is unavailable, so findings are downgraded and verification is marked unavailable, never fabricated. Demonstrated as one full benchmark pass with the model server off, showing which capabilities survive and which numbers drop.